

KnowSelf: Teaching LLM Agents

When to Think, Reflect, or Seek Knowledge

Agentic knowledgeable self-awareness for interactive planning (ALFWorld)

Source: arxiv.org/abs/2504.03553

Fast / Slow / Knowledgeable

Selective knowledge use

Efficiency + accuracy

Current Agent Training Uses "Flood Irrigation" — Indiscriminate Knowledge Injection

Existing methods inject gold trajectories, external feedback, or domain knowledge regardless of situational need → fragile planning + unnecessary

Trajectory imitation

(1) Direct trajectory imitation (e.g., ReAct)

Learns patterns from gold trajectories

Can collapse on unexpected inputs

Human analogy: situational self-awareness

Act directly vs reflect vs seek external help

KnowSelf brings this metacognition to LLM agents

Trial-and-error

(2) Trial-and-error refinement (e.g., Reflexion)

Uses external feedback to refine

Still not situationally selective

Knowledge-augmented

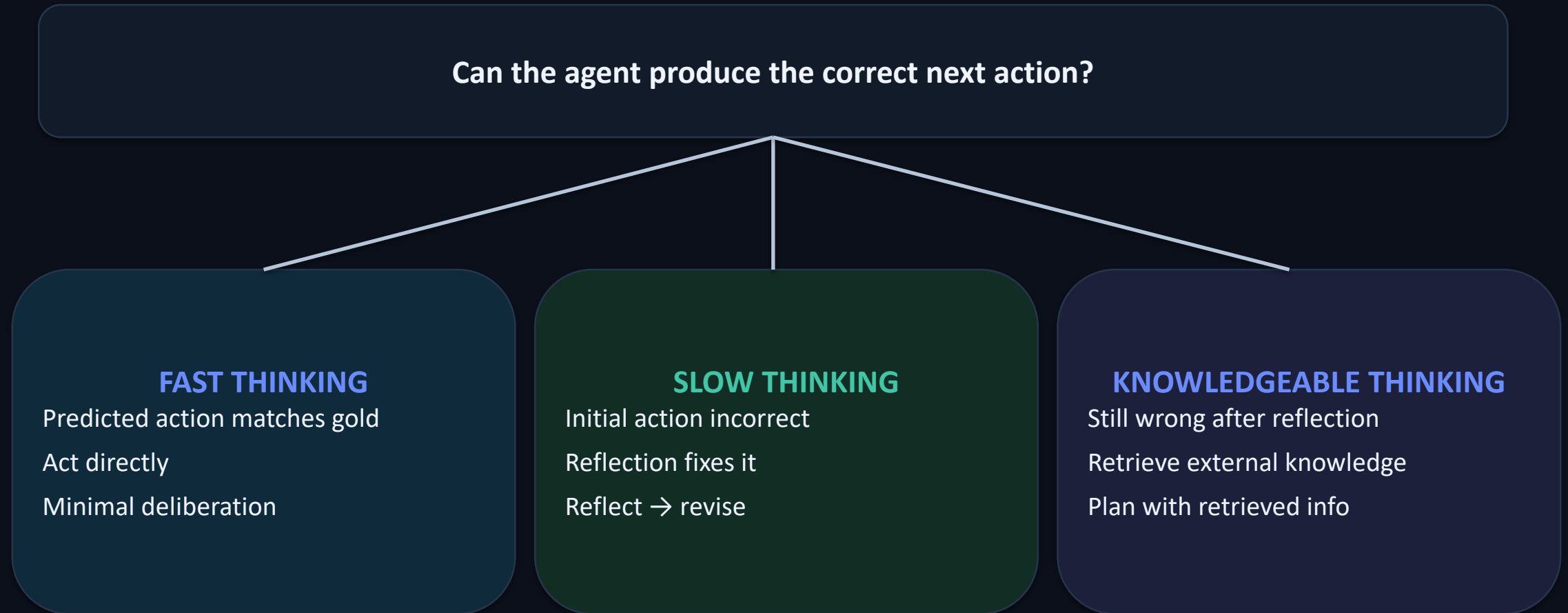
(3) Knowledge-augmented planning (e.g., KnowAgent, WKM)

Indiscriminate domain-knowledge injection

Often 100% knowledge steps → high cost

Three Situational Modes: Fast Thinking, Slow Thinking, Knowledgeable Thinking

Criterion per step: direct correct action → reflect-correctable → requires external knowledge (framework figure)



KnowSelf Framework: Data Construction → Two-Stage Training → Self-Aware Inference

Step 1 — Self-awareness data construction

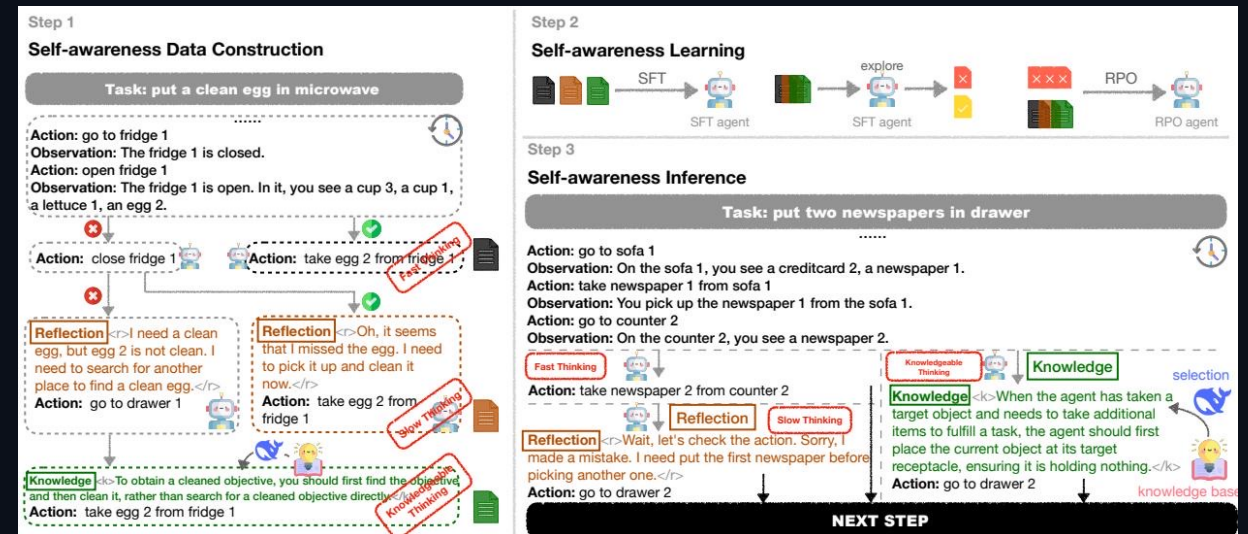
- self-explore environments
- heuristic situation judgment per step
- insert special tokens for fast/slow/knowledgeable

Step 2 — Two-stage training

- SFT then RPO

Step 3 — Self-aware inference

- generate tokens to trigger action / reflection / retrieval



Framework overview (paper figure)

KnowSelf Correctly Heats a Mug and Places a Saltshaker Where Baselines Fail

Qualitative comparison from the paper (figure)

(a) Task: put a hot mug in cabinet	
..... Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. [Knowledge]<k>If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of placing the object on it.</k> Action: take egg 1 from microwave 1 Observation: Nothing happens. 	 Action: take mug 1 from countertop 1 Observation: You pick up the mug 1 from the countertop 1. Action: go to microwave 1 Observation: The microwave 1 is closed. Action: open microwave 1 Observation: You open the microwave 1. In it, you see an egg 1. Action: put mug 1 in/on microwave 1 [Reflection]<r>Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.</r> Action: heat mug 1 with microwave 1
(b) Task: put a saltshaker in drawer	
..... Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is open. In it, you see a fork 1. Action: put saltshaker 1 in/on drawer 1 [Reflection]<r>Wait, the drawer is open, so the saltshaker might fall inside. I need to close the drawer first.</r> Action: close drawer 1 Observation: You close the drawer 1. Action: put saltshaker 1 in/on drawer 1 Observation: Nothing happens. 	 Action: take saltshaker 1 from countertop 3 Observation: You pick up the saltshaker 1 from the countertop 3. Action: go to drawer 1 Observation: The drawer 1 is closed. Action: open drawer 1 Observation: You open the drawer 1. In it, you see a fork 1. [Knowledge]<k>When the agent encounters a situation where receptacles contain other items but only needs to place the target object, the agent should proceed to place the object without removing unrelated items.</k> Action: put saltshaker 1 in/on drawer 1

Put a saltshaker in drawer: KnowSelf applies a knowledge token to place the object without removing unrelated items; DeepSeek-R1 gets stuck.

KnowSelf Achieves 84.33% on ALFWorld (Llama-8B) Using Only 15% Knowledge

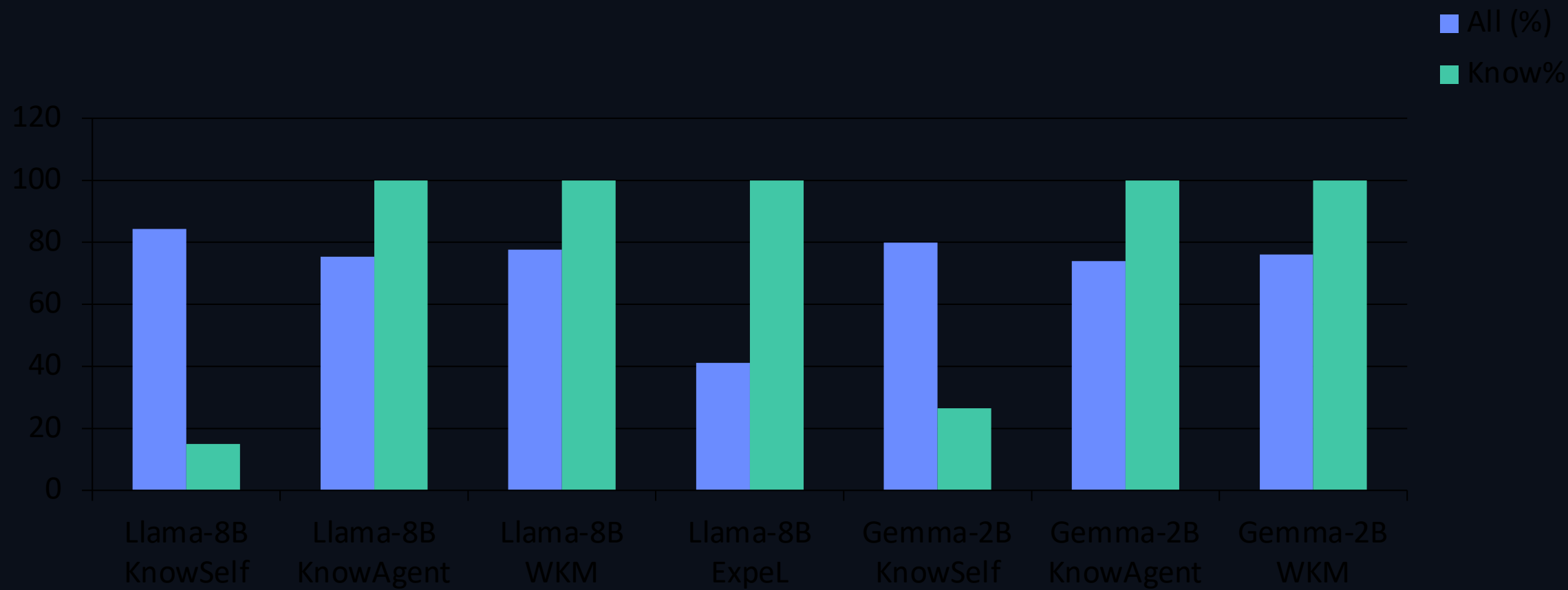
Success rates (%) across 6 task types; Know% = proportion of steps using external knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

KnowSelf uses far less knowledge (15.01% / 26.41%) than 100%-knowledge baselines and often performs better.

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage

Bar chart contrasts Know% (~15–26% vs 100%) against All accuracy across methods and model scales



KnowSelf Know%: 15.01 / 26.41

Baselines Know%: 100

Accuracy stays high

Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

Indiscriminate knowledge injection is wasteful and can be suboptimal.

Agents can learn situational self-awareness via special tokens with lightweight heuristic labeling.

KnowSelf (Llama-8B) achieves 84.33% All, outperforming GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%).

Only 15.01–26.41% knowledge usage needed vs 100% for ExpeL / KnowAgent / WKM.

Generalizes across model scales: Gemma-2B + KnowSelf reaches 79.85%.